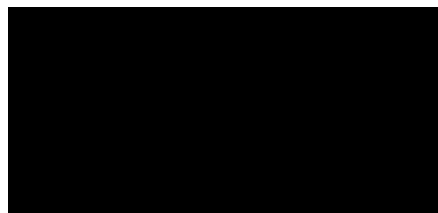


University of Waterloo
Faculty of Engineering
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Detailed Design and Project Timeline

Educational RF Receiver
Group 2017.008



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1 HIGH-LEVEL DESCRIPTION OF PROJECT

1.1 Motivation

Over the past two decades, wireless communication has seen drastic advancements, through the introduction of sophisticated communication standards in IEEE 802 [1]. These improvements call for stringent requirements on radio frequency (RF) circuits and systems, and emphasize the importance of detailed noise and distortion analysis on radio systems. In RF circuits and communication systems courses, students are required to analyze intrasystem signals examine the impacts of each building block on the overall performance. Unfortunately, the existing equipment in laboratories does not allow the measurement of these signals.

For example, the communication systems course at the University of Waterloo (ECE 318) uses the TIMS 301/C system unit to teach students. This system, while modular, only operates at a maximum frequency of 100kHz [2]. This does not allow students to experience much of the challenges of designing a true high frequency RF system.

Generally, RF equipment is extremely expensive due to its high sensitivity to external effects. For an undergraduate class where the university might need to purchase 20 or 30 units of the equipment, it is difficult for the university to set aside a large budget for equipment that is only used for training.

1.2 Project Objective

The objective of this project is to design a reconfigurable RF receiver. The receiver takes in high frequency RF signal, converts it down to lower frequencies and demodulates the analog signal for digital signal post-processing and visualization. Students can observe the effects on signals at multiple stages of signal processing by adjusting additive noise, filters, and in-phase and quadrature (IQ) components. Students can also measure several figures of merit, such as noise figure, error vector magnitude, gain and distortion at different stages of the receiver.

1.3 Block Diagram

The high level design of the reconfigurable receiver module is described in Figure 1. The system can be divided into four major subsystems under the overall system: Radio Frequency (RF), Intermediate Frequency (IF), baseband, and software. The RF front-end consists of all components between receiver RF input and the baseband and includes the RF and IF subsystems. The RF subsystem is in the same frequency domain as the received signal between 0.8 GHz and 1.9 GHz. The signal is then converted down to the IF frequency domain at $140\text{MHz} \pm 10\text{MHz}$. This frequency conversion down to the IF frequency domain reflects common industry heterodyne receiver architecture. In the baseband system, the signal is then converted down to baseband frequencies to be demodulated and decoded. The baseband system then forwards the signal onwards to the software subsystem, which stores the data and presents it in a manner that is pleasing and informative to the user.

In addition to the four major subsystems, the receiver also requires a power distribution subsystem as well as a RF front-end control subsystem. The power subsystem is responsible for supplying power to all of the components in the RF, IF and baseband subsystems. It is important that the input power to each of the components is clean and meets the component specification requirements. This is particularly important for RF and IF subsystem components. The RF front-end control subsystem delivers control signals from the user interface as well as signals required to power on the receiver from a power cycle state. Signal outputs are located at four-receiver signal stage shown in Figure 1. At each connection point of the receiver, a directional coupler is used to split the input signal between the Sub-Miniature version A (SMA) connector and the next stage in the receiver. A SMA connector is connected between every stage of the receiver to allow end users to connect test equipment and observe the signal.

An in-depth system diagram of the software used to control and interface with the receiver is described in Figure 2.

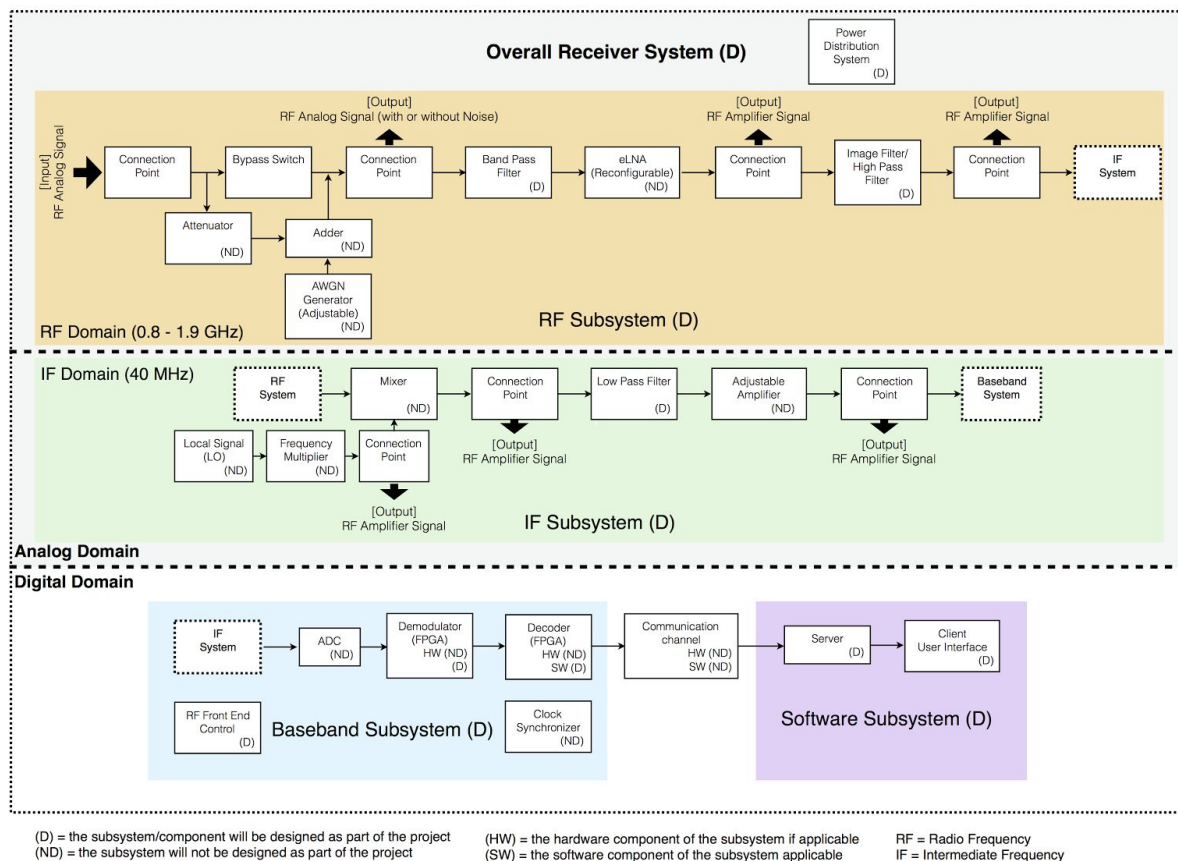


Figure 1. Block Diagram of the Educational Radio Frequency Receiver

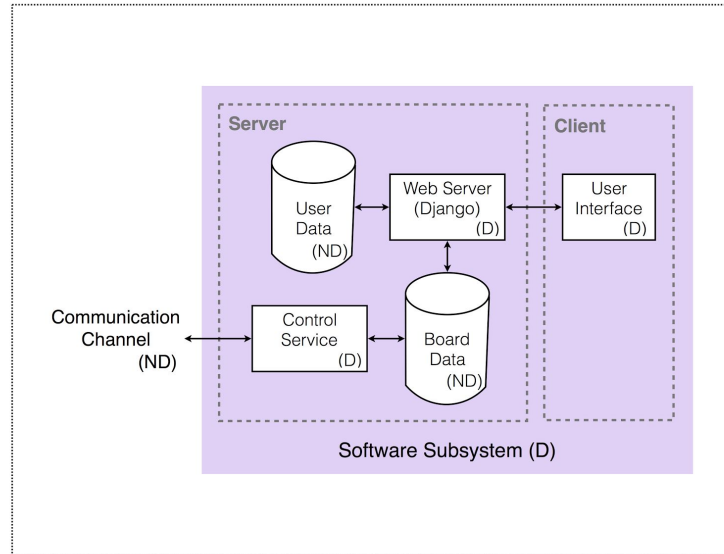


Figure 2. In-depth Block Diagram of Software System

2 PROJECT SPECIFICATIONS

2.1 Functional Specifications

Functional specifications list criteria for specific behaviors that the receiver must demonstrate. These behaviors provide educational value to the receiver such that its settings can be changed and students are given insight into how these changes impact the various subsystems of the receiver. Table 1 lists essential functional specifications while Table 2 lists non-essential functional specifications.

Table 1: Essential Functional Specifications

Specification	Subsystem	Description
Support carrier frequencies between 0.8 and 1.9 GHz	Overall Receiver System	In order to receive a signal with different carrier frequencies between 0.8 and 1.9 GHz, the receiver must adjust the settings of RF and IF subsystem components. These settings will be adjusted through the user interface.
Support a maximum bandwidth of 20 MHz	Overall Receiver System	Receiver supports up to 20 MHz bandwidth, which is the current long term evolution (LTE) single carrier maximum bandwidth.
Support AM (amplitude modulation), BPSK (binary phase-shift keying), QPSK (quadrature phase-shift keying), and 16 QAM (16	Baseband	The receiver must be capable of demodulating and decoding common industry modulation schemes. This capability is important for students studying communication systems to understand different modulation schemes.

quadrature amplitude modulation) modulation schemes		
Overvoltage/overcurrent power protection	Overall Receiver System	If either input voltage or current is greater than 150% of expected values for more than 100 ms, the power protection system is required to activate.
Sensitivity is greater than -70 dBm	Overall Receiver System	Receiver must be able to receive and accurately decode signals with power greater than -70 dBm at which the decode bit error rate must be less than 0.1%.
Capable of attenuating signal by 30 dB to 70 dB	RF	Attenuator capable of attenuating signal from by 30 to 70 dB is placed as first component in receiver architecture. Attenuator is placed to simulate loss in wireless communication.
Directional coupler directs -10 dB $\pm$ 3 dB of input power to SMA to measure noise figure, gain, distortion, etc.	Overall Receiver System	To measure the signal at each access point, probing the RF signal trace will not provide accurate measurement. In radio frequency, a probe is considered as extra transmission line attached to the system, which changes impedance, phase and reflection of signal. For accurate measurement of radio frequency signal, directional coupler is placed at every probing access point to sample RF signal, without providing changes to entire system. Directional coupler transfers -10 $\pm$ 3 dB of input signal to SMA.
Support the connection of a noise generator to simulate additive noise	RF	An SMA connector will provide the ability to add noise to received signal for analysis of receiver performance of noisy signals.
Less than 3 s of latency when modifying settings from user interface	Software	Any RF hardware setting change must be capable of being reflected in receiver operation in less than 3 s after modifying setting through client application. Client application is using web page through internet, which expose to further latency than local data transfer.

Table 2: Non-Essential Functional Specifications

Specification	Subsystem	Description
Wireless Operation through SMA Antenna	RF	Antennas with SMA connections are available. By plugging in an antenna and closing the bypass switch, the board should be able to sniff wireless packets from

		over-the-air.
Multiple Account Access Privileges	Software	Application will be able to store 120-account information with private file storage system. The application will establish two types of accounts: instructor and student with links between the two.

2.2 Non-functional Specifications

In addition to the functional specifications, the project also contains several non-functional specifications that are outlined below. Table 3 lists eight essential non-functional specifications while Table 4 lists five non-essential non-functional specifications.

Table 3: Essential Non-Functional Specifications

Specification	Subsystem	Description
Web application interface must support latest Safari, Google Chrome, and Firefox Web Browser	Software	As a web application, the students will be able to access the software from a variety of devices and will not limit students to a particular operating system or interface.
All software features and receiver settings must be accessible within 4 clicks	Software	In order to provide a low learning curve to students starting out with the application, the goal is to provide an interface with receiver settings and received data readily accessible to the user.
RF front-end component cost must not exceed \$500	RF	RF front-end component cannot be easily changed or swapped without going through a rigorous redesign or requalification process. It is important that the RF front-end component cost is relatively low in order to create an affordable product.
Cooling sufficient for at least 120 minutes of max power operation	Overall Receiver System	Receiver should be able to operate for at least 120 minutes at max power without encountering any stability or cooling issues.
Power Consumption less than 10 W	Overall Receiver System	The maximum use-case power consumption by all RF front-end components will not exceed 10 W.
Case Size is insulated and under 50x50x20 cm ³	Overall Receiver System	All the circuitry of the receiver with the exception of connectors must fit within a casing smaller than 50x50x20 cm ³ . This is required so that the receiver takes up no more space than 50% of the standard laboratory desk. The case will be insulated for greater

		than 120 V to protect users.
Web server must support at least 30 simultaneous users	Software	Thirty is a realistic number of groups in a RF lab session. Every group should be able to quickly access/store lab data and waveforms and upload/download documents at all times during a lab session.

Table 4: Non-Essential Non-Functional Specifications

Specification	Subsystem	Description
Weight of receiver is under 5 kg	Overall System	Receiver should be light enough for an individual to carry and move the equipment from storage area to a laboratory in the same building.

3 DETAILED DESIGN

3.1 Subsystem 1 Design: RF/IF Subsystem

3.1.1 Overview of RF/IF Subsystem

The goal of RF and IF subsystems are to ensure clear signal is extracted and passed onto baseband subsystem to decode. For educational purposes the RF and IF subsystem components are broken down into stages for observing signal process changes. The subsystems include components such as filters, eLNA, mixer, and amplifiers to ensure signal integrity.

To explain the need for various analog components in RF and IF subsystem, signal processing details are examined as signal pass through each stage. Immediately after RF analog signal is received, a bypass switch is in place to manually control signal flow. Since the system might demonstrate with input signal directly from signal generator, attenuation and noise need to be added to mimic real life signal interference. A band pass filter is followed to filter out any signal outside of interested frequencies (800MHz to 1.9GHz) as well as avoid overloading external linear amplifier (eLNA). At high frequency of this project, any type of amplifier including eLNA will produce nonlinear distortion at output signal which produces sinusoidal harmonics of the input signal. Equation 1 shows distortion generation in eLNA by modeling amplifier's transfer characteristic with a simple power series function, the nonlinear component of output voltage signal is indicated as the third and fifth harmonics of output voltage.[3]

$$V_{out} = a_1 V_{in} - a_3 V_{in}^3 - a_5 V_{in}^5 - \dots \quad (1)$$

As the last component of RF subsystem, an image rejection filter is required to remove the harmonic components and distortions created by eLNA. This filter is very essential to the IF subsystem and is limited by the IF subsystem frequency. In order to demodulate the signal in baseband subsystem the

RF signal needs to be downconvert to an intermediate frequency of 40MHz. The 40MHz frequency requirement is determined by the analog-to-digital converter (ADC) in baseband subsystem as it can not process signal faster than 40 MHz. This puts constraint on the image filter to provide high rejection and sharp cut off edges.

The image filter protects mixer in IF subsystem from downconverting unwanted noise or distortion to 40MHz. As the input signal $m(t)$ mix with a carrier signal $c(t)$ (equation 2) the output signal $u(t)$ (equation 3) contains signal at two frequencies $f-f_c$ and $f+f_c$. Of the two frequencies the $f-f_c$ will be centered at 40MHz and will pass through the low pass filter in IF subsystem. Figure 3 shows the location of RF and IF signals on a frequency spectrum. [4] The LO- RF frequency represent the 40MHz signal which is what the original input signal is.

$$c(t) = A_c \cos(2\pi f_c t) \quad (2)$$

$$\begin{aligned} u(t) &= A_c * m(t) \cos(2\pi f_c t) \\ &= (A_c/2) * m(t) e^{j2\pi f_c t} + (A_c/2) * m(t) e^{-j2\pi f_c t} \\ &= (A_c/2) * m(f-f_c) + (A_c/2) * m(f+f_c) \end{aligned} \quad (3)$$

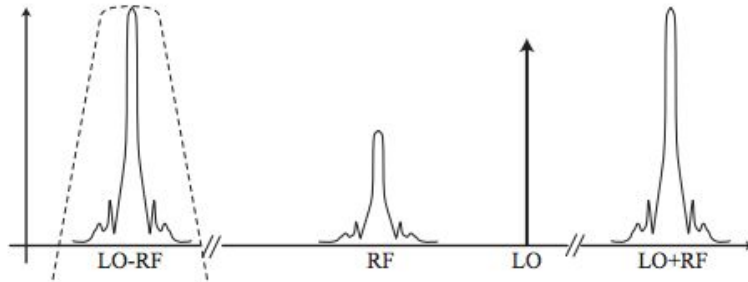


Figure 3. IF subsystem frequency illustration [4]

As the mixer convert RF signal down to LO-RF, signal at other frequency can also be mixed in. As shown in Figure 4, both frequency LO+IF and LO-IF is mixed with frequency LO to the output IF frequency. Since we only want signal at LO+IF frequency, image rejection filter is placed to reject the LO-IF frequency.

For example:

$$\begin{aligned} \text{Require : } IF &= 40\text{MHz} \\ \text{Assume : } RF &= LO + IF = 1.9\text{GHz} \\ LO &= 1.9\text{GHz} - 40\text{MHz} = 1.86\text{GHz} \\ LO - IF &= 1.96\text{GHz} - 40\text{MHz} = 1.82\text{GHz} \\ \Delta &= LO + IF - (LO - IF) = 2 * IF = 80\text{MHz} \end{aligned}$$

The difference between the signal frequency (LO+IF) and distorted frequency (LO-IF) is only 80MHz meaning the filter needs a very sharp and clear cut on the edges to eliminate the undesirable distortion (LO-IF). [5] Further details of filter selection and design will be discussed in section 3.1.2.

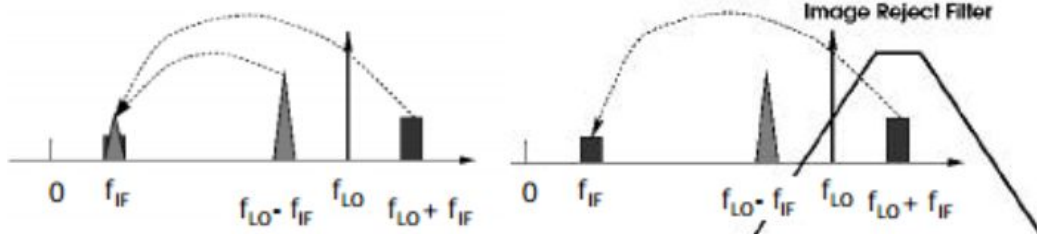


Figure 4. Image Rejection Filter illustration [5]

The variable gain amplifier (VGA) is the last stage of the IF system after low pass filter. Since eLNA in RF subsystem can't amplifier signal to full range of ADC input voltage swing, due to the high frequency, the variable gain amplifier at IF subsystem will boost the signal to full range.

3.1.2 Image Filter Design

3.1.2.1 Image filter Parameters

There are 3 main performance parameters to consider for image filter selection.

1) Bandwidth

Filter bandwidth should be greater than signal bandwidth, 20 MHz.

2) Insertion loss at signal passband frequency

$$\text{Insertion Loss(dB)} = -20 \log |S_{21}| \quad (4)$$

Insertion loss, defined in equation 4, quantifies how much power is lost when signal pass through the filter. Insertion loss at passband should not exceed 3dB, to ensure more than 50% of input power transfer to output port.

3) Attenuation at image frequency

Attenuation at image frequency is calculated using equation 4 (insertion loss outside of passband is called attenuation). Attenuation at image frequency should be greater than 33dB to ensure that more than 99.95% of image signal is attenuated in filter.

$$\% \text{ of Power Attenuated} = 100\% * (1 - 10^{-33\text{dB}/10}) = 99.95\%$$

For optimal filter selection, off-the-shelf filter and custom design filter are compared and analyzed. Operation point of filters used in analysis are as follows: center frequency at 1.9GHz, passband frequency from 1.89GHz to 1.91GHz, image frequency at 1.82GHz.

3.1.2.2 Off-The-Shelf Image Filter

Saw filter AFS14A40-1900.00-T3, from ABRACON Corp. is chosen as off the shelf filter option.



Figure 5. S21 response of off-the-shelf image filter [6]

From Figure 5, bandwidth of this filter is 40MHz, covering passband frequency (1.89GHz to 1.91GHz). Average insertion loss at passband is 2dB, and attenuation at image frequency, 1.82GHz, is 36dB. [6]

$$\text{At passband: } Power_{out}/Power_{in} = 10^{-2dB/10} = 0.631$$

$$\text{At image frequency: } Power_{out}/Power_{in} = 10^{-36dB/10} = 0.000251$$

This filter passes 63.1% of input signal at passband, while attenuating 99.9749% of input signal at image frequency. This clearly satisfies the filter requirements, discussed in section 3.1.2.1.

3.1.2.3 Custom Design Image Filter

(1) Lumped Element Filter

First option in custom design is lumped element filter, which uses inductors, capacitors, and resistors. Figure 6 shows filter prototype at 1.9GHz and its response is presented in Figure 7.

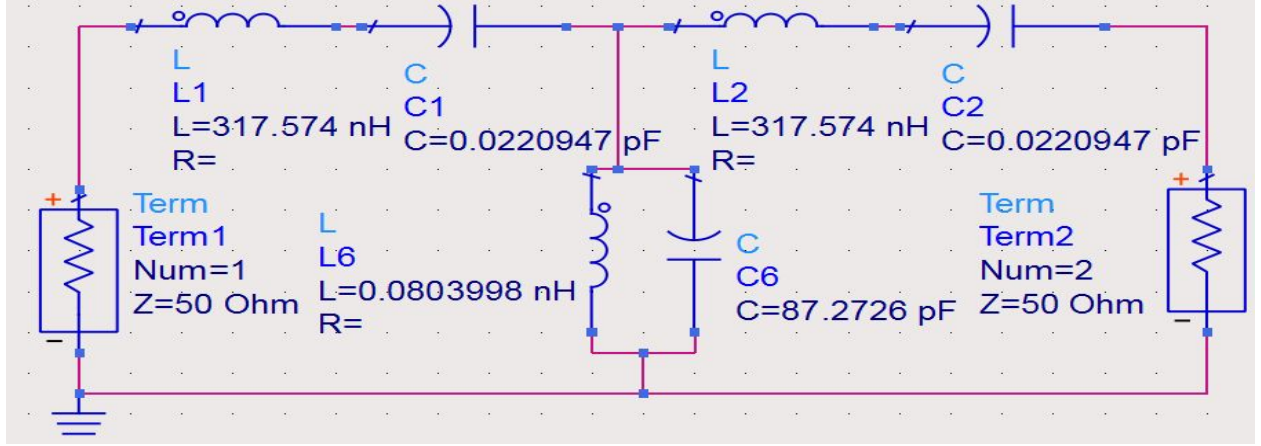


Figure 6. Lumped Element Filter Prototype with ideal components

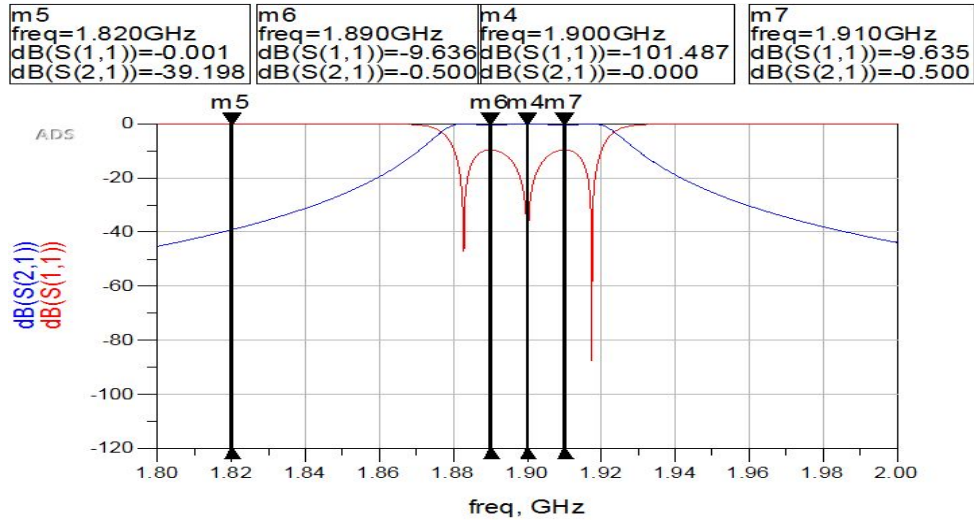


Figure 7. Lumped Element Filter Simulation Response

Simulation result from design with ideal components is very promising. Bandwidth is 40MHz, covering passband from 1.89GHz to 1.91GHz. Insertion loss at passband is less than 0.5dB and attenuation at image frequency, 1.82GHz is 39.198dB. All filter parameters meet design requirement, discussed in section 3.1.2.1.

However, there are two non-ideal factors to consider in real RF circuit board. First factor is parasitic effect. Parasitic resistance, parasitic capacitance, and parasitic inductance must be considered at high frequency. Parasitic resistance is quantified by quality factor(Q), expressed in equation 5 and 6.

$$Q_{capacitor} = 1/(C * 2\pi f * R_{Parasitic}) \quad (5)$$

$$Q_{inductor} = L * 2\pi f / R_{Parasitic} \quad (6)$$

Parasitic resistance results in loss in signal path, increasing insertion loss. Furthermore, parasitic inductance and parasitic capacitance shifts frequency response. Figure 8 shows simulation result

when ideal inductors and capacitors are replaced with non-ideal components, simulating parasitic effect.

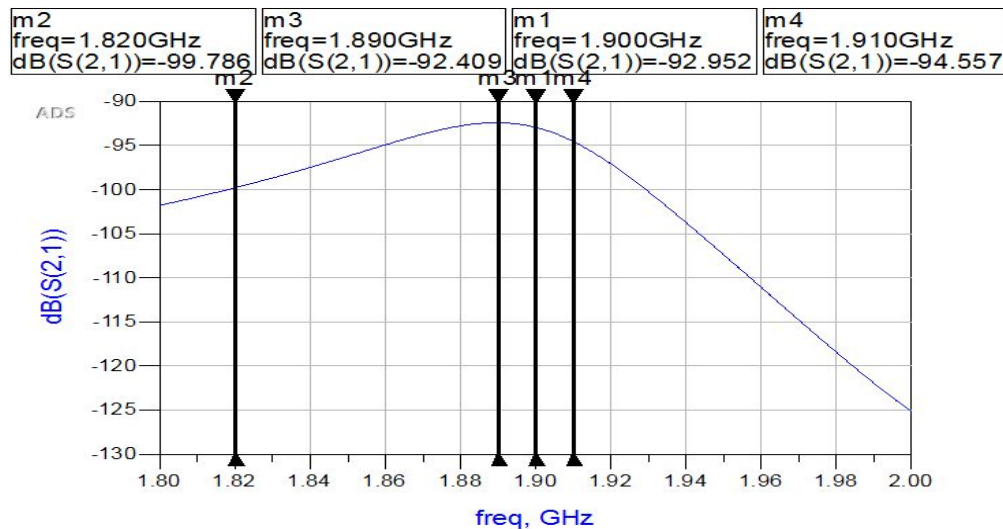


Figure 8. Lumped Element Filter Simulation Result with parasitic effects.

In Figure 8, insertion loss is significantly increased to 95dB. Frequency response is shifted by 10MHz to the left. Design requirements is no longer satisfied in lumped element filter.

Second non-ideal factor is capacitor and inductor values. Capacitor and inductor values required in the design are too low. There are no suppliers who support 0.02nF capacitor or 0.08nH inductor. Minimum value available is 0.1nH inductor and 0.1pF capacitor. Theoretically, it is possible to implement 0.08nH inductance and 0.02nF capacitance, by putting large value elements in parallel or in series. However, increasing number of components in board will increase insertion loss due to parasitic resistance.

(2) Coupled Line filter

To overcome limitation on lumped element filter, coupled line filter is used as second custom design option. Instead of using lumped elements, it uses two adjacent transmission lines. Crosstalk between closely spaced transmission line is used in the design to pass signal only in passband frequency. Length, width and separation of two adjacent transmission lines are calculated to set characteristic impedance and coupling factor.

Custom design coupled line filter layout is shown in Figure 9 and its response is presented in Figure 10.

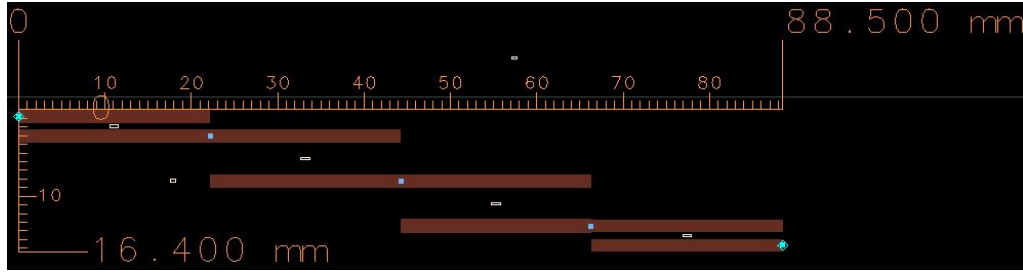


Figure 9. Coupled Line Filter Layout using FR4 as dielectric substrate

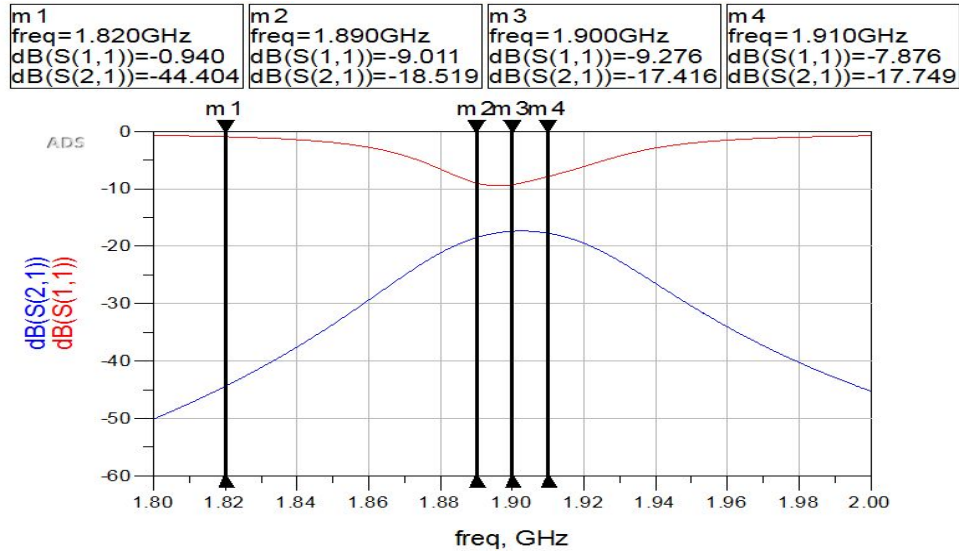


Figure 10. Coupled Line Filter Simulation Result

Bandwidth of coupled line filter is from 1885MHz to 1915MHz, covering passband frequency. Attenuation at image frequency is 44.4dB, which satisfies design requirement. However, insertion loss at passband is 17.4dB, which fails design requirement. Furthermore, filter size is 88.5mm by 16.4mm. Adding coupled line filter in design increases total board size. Having bulky filter in the final board, not only increase implementation cost, but also make routing and placement of PCB complicated.

3.1.2.4 Image Filter Selection

Custom design filter has limitation on its size and performance. Although higher attenuation at image frequency is achieved in custom design filter, custom design filter can not meet insertion loss design requirement. Furthermore, custom design filter requires 88.5mm by 16.4mm area, making PCB routing and placement complex. Off-the-shelf component meets all design requirement, and requires only 1.1mm by 1.4mm area. From this analysis, off-the-shelf filter is chosen as final image filter in design.

3.1.3 Gain Stage Design

The design goal for the gain stage is to ensure that a sufficient amount of gain is present to amplify the receiver input while preserving a high of a signal-to-noise ratio (SNR) as possible.

The required gain for the entire RF system is dependent on the sensitivity of the receiver (weakest RF signal for which it's designed) and the full-scale range of the ADC (necessary for maximum precision). The ADC candidate for the receiver is the ADS4146 chip from TI. From the datasheet [7], the full-scale range of its input is 2V peak-to-peak (Vpp). The characteristic impedance is $200\ \Omega$ from the output of the mixer candidate ADRF6601 [8]. Using equation 7 [8], the required power at the ADC is $3.979\text{dBm} \approx 4\text{ dBm}$.

$$dBm = dBmV - 10\log_{10}\left(\frac{R}{1*10^{-3}}\right) \quad (7)$$

The components between the receiver RF input and the ADC are: bypass switch, bandpass filter, eLNA, image filter, mixer, low pass filter, adjustable amplifier and 4 output connection points. Of these components, the components that have a $|dBgain|$ greater than 3dB are the eLNA, mixer and adjustable amplifier. To simplify analysis, only these 3 components will be considered in the gain analysis. The gain for these three components are shown in Table 5.

Table 5. Component Gains

Component	Part Number	Gain (dB)
LNA	ADL5521	15.8 [9]
Active mixer	ADRF6601	6.7 [8]
Adjustable amplifier	ADRF6518	66 [10]

The total cascaded gain is 88.5dB. In order to have the maximum performance at the input power of -70dBm, 4 dBm of power must be present at the ADC; thus, the total cascaded gain must be 74dB. The VGA can adjust its gain between 66dB and -36dB so because 88.5dB is greater than 74dB, the gain can be adjusted to achieve full-scale range while not clipping the ADC. Thus, this gain stage design will have sufficient gain for maximum precision when receiving a signal as low as -70dBm.

An important equation when trying to maximize SNR ratio is the Friis equation (equation 8). The Friis equation determines the total cascaded noise figure (degradation of SNR) from the noise figures and gain of individual components [9]. F is the total noise figure of the entire system while F_x and G_x are the noise figure and gain of component x.

$$F = F_1 + \frac{F_2-1}{G_1} + \frac{F_3-1}{G_1G_2} + \frac{F_4-1}{G_1G_2G_3} + \dots \quad (8)$$

The thermal noise floor at room temperature is -101dBm [9] for 20 MHz signal bandwidth. For a -70dBm signal, the initial SNR ratio is 31dB. In section 3.2.4, it is discovered that a minimum SNR ratio of 20dB is required for the demodulator to lock onto the carrier frequency. Thus the total noise figure must be less than 11dB.

The reason that the LNA and the adjustable amplifier are not both placed before the mixer is because the mixer will begin to compress the output at higher input powers. However, a clear design decision was made to place the adjustable amplifier with high gain later in the gain stage while placing the LNA with a low noise figure at the beginning of the gain stage.

A separate amplifier that was considered was the RF3374 chip. A comparison between this chip and the ADL5521 LNA is demonstrated in Table 6 along with the parameters of the active mixer, adjustable amplifier and SAW filter.

Table 6. Component Noise Figure and Gain.

Component	Part Number	Gain (dB)	Noise Figure (dB)
LNA	ADL5521	15.8 [10]	1.0 [10]
Amplifier	RF3374	18.9 [12]	3.5 [12]
SAW filter	B5094	-2.1	2.1
Active mixer	ADRF6601	6.7 [8]	14.8 [8]
Adjustable amplifier	ADRF6518	-36 ~ 66 [11]	-65.4 at 51.5 dB gain 22.1 at min gain

The gain of the SAW filter is the inverse of the insertion loss which is 2.1 dB [6]. The noise figure of the SAW filter is found using equation 9 where A is attenuation and F is noise figure for that component [13].

$$F = A \quad (9)$$

The noise figure of the adjustable amplifier is found using equation 10 [10]. Where N_{out} is the noise density of the amplifier and $N_{50\Omega}$ is the noise density of a 50Ω resistor at room temperature.

$$F = 10 \log_{10} \frac{(N_{out}/GAIN)}{N_{50\Omega}} \quad (10)$$

The noise figure of the adjustable amplifier varies inversely proportionally with the gain. However, the required gain of the adjustable amplifier to reach full-scale range is also inversely proportional to the received signal strength. Thus, the noise figure of the adjustable amplifier varies proportionally with the received signal strength. As a higher received signal strength results in a higher SNR. As a result, the overall impact of the adjustable amplifier on the system SNR delivered to the ADC should remain

the same for all received inputs and it is necessary to only calculate it at one receive power and assuming that it is constant.

The system noise figure calculation for ADL5521 and RF3374 is shown in Table 7. Note, the system noise figure calculation is for a received signal power of -70dBm where the adjustable gain is 51.5dB.

Table 7. System Noise Figure Calculation.

First Stage Amplifier	F_1 (W/W)	$\frac{F_2-1}{G_1}$ (W/W)	$\frac{F_3-1}{G_1G_2}$ (W/W)	$\frac{F_4-1}{G_1G_2G_3}$ (W/W)	F (dB)
ADL5521	1.26	0.02	1.26	~0	4.05
RF3374	2.24	0.01	0.62	~0	4.58

From Table 7, the use of ADL5521 gives a lower system noise figure which results in a higher SNR at the ADC input. Thus, ADL5521 should be used. For a received signal power of -70dBm, the input SNR is 31dB. The required SNR for the demodulator to lock is 20 dB. Thus, there remains 7 dB of headroom.

3.2 Subsystem 2 Design: Baseband Subsystem

A detailed block diagram for the baseband subsystem is illustrated in Figure 11. As it can be observed, everything inside the box labeled FPGA, which consists of elements used for the demodulator, is implemented in the FPGA. The signal generator and sampler serve the purpose of generating the input for the demodulator. First, the signal generator passes a signal with a bandwidth of 20 MHz centered around the intermediate frequency (IF) as input, which then goes through the sampler, which is the simulated analog-to-digital converter.

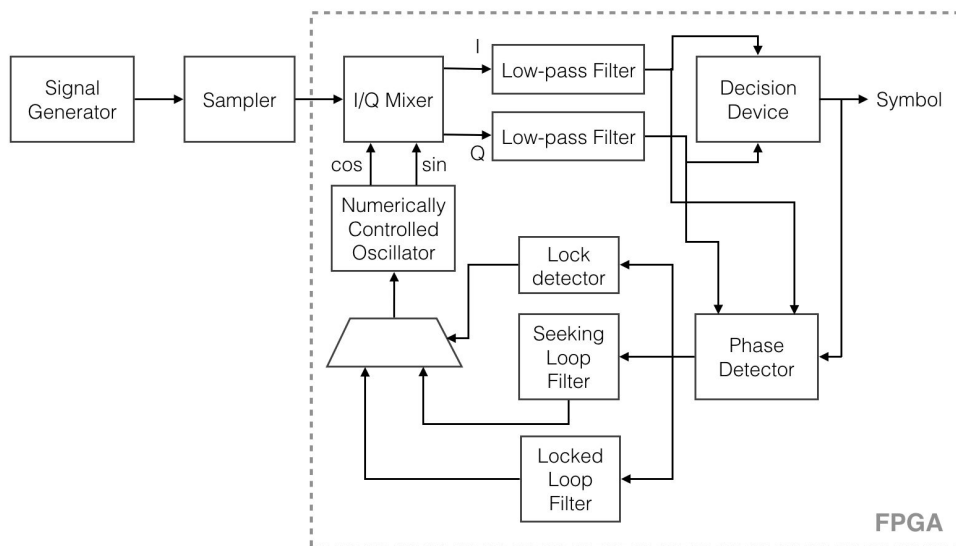


Figure 11. Detailed block diagram for the baseband subsystem.

This signal is expected to have a frequency error with an upper bound of 1000 ppm. Current crystal oscillators have an upper bound on the error rate of 10 ppm. The maximum frequency at which signals are received is 1.9 GHz, and it can be downmixed to an intermediate frequency of 40 MHz. Therefore, by heterodyne principle, the maximum frequency that can be generated by the local oscillator for downmixing is thus 1.9 GHz + 40 MHz = 1.94 GHz.

$$(1.9 \times 10^9 \text{ Hz}) \times 10 \times 10^{-6} = 19 \times 10^3 \text{ Hz} \quad (11)$$

$$(1.94 \times 10^9 \text{ Hz}) \times 10 \times 10^{-6} = 19.4 \times 10^3 \text{ Hz} \quad (12)$$

$$\frac{38.4 \text{ kHz}}{40 \text{ MHz}} = 960 \text{ ppm} \approx 1000 \text{ ppm} \quad (13)$$

Adding equations 11 and 12, an error value of 38.4 kHz is obtained, and dividing that by the intermediate frequency (f_{IF}) gives us the upper bound for the input signal, which is rounded to 1000 ppm to provide some additional headroom in design, as shown in equation 13.

As the signal leaves the sampler, which samples the signal at 160 MSPS, it goes into the in-phase (I) and quadrature (Q) mixer, which separates the in-phase and quadrature components. Then, the signal gets filtered using a low-pass filter to obtain the low frequency components, which is then used by the decision device to determine the symbols in the message. The ideal symbol locations, which is called the constellation, is shown in Figure 12.

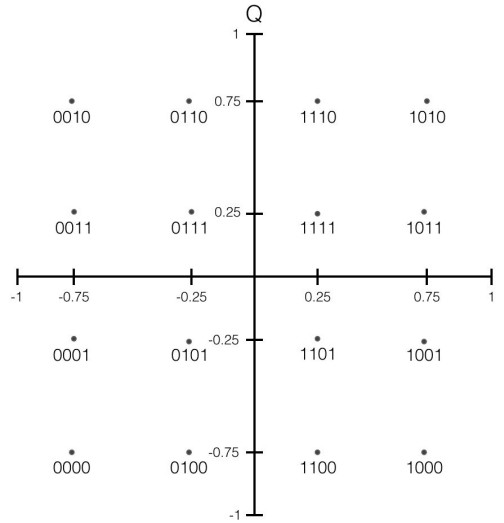


Figure 12. 16 QAM constellation

An issue that arises with constellations is caused by phase and frequency errors, as depicted in Figure 13. Phase errors produce a static rotation that is time-invariant on the constellation, whereas frequency errors rotate the constellation with time, causing it to spin.

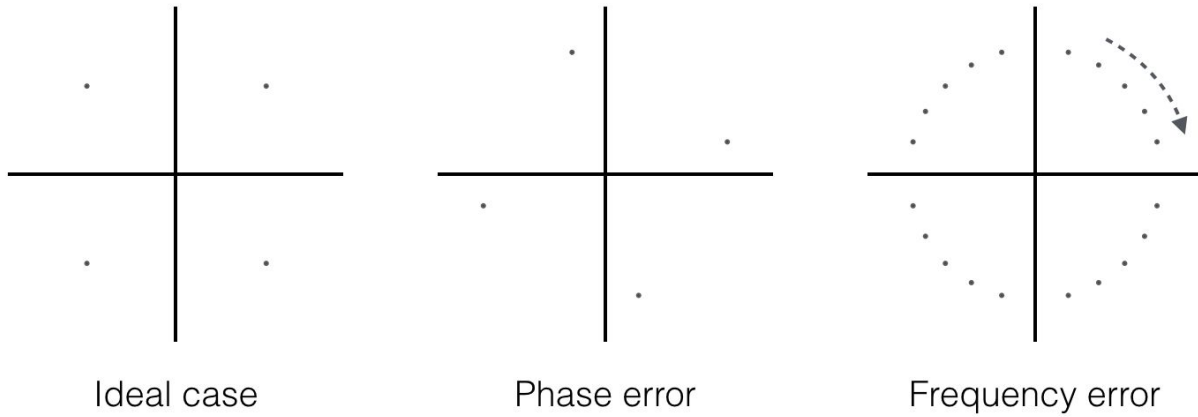


Figure 13. QPSK constellation showing the ideal case, with phase error, and with frequency error

In order to solve these errors, the design of the demodulator makes use of a negative feedback loop comprised of a low-pass filter, a decision device, a phase detector, and a loop filter which are explained in the following sections. The process of correcting the phase and frequency will be referred from hereon out as phase correction, since the detector cannot actually detect frequency errors and instead correct for them as a time varying phase error. The importance of phase correction is illustrated in Figure 14, where one can observe that the constellations are completely undecipherable since they spin with time.

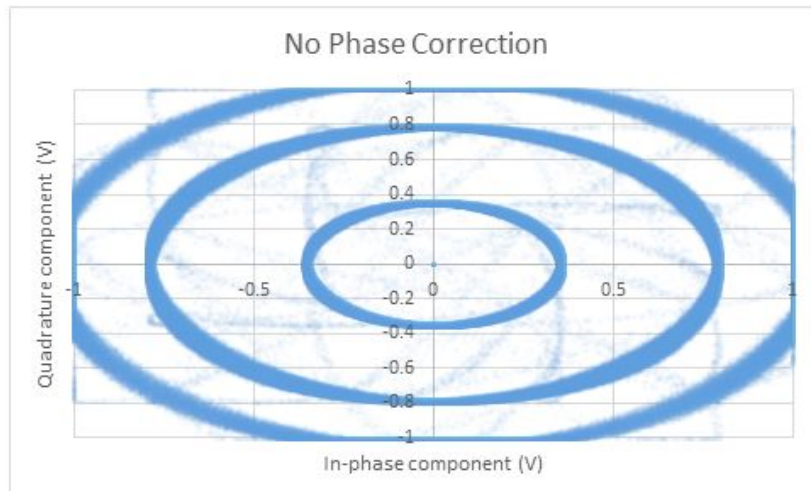


Figure 14. Constellation for no phase correction

3.2.1 ADC sample rate

As the signal goes through the sampler, sampling rate needs to be determined. Since $f_{IF} = 40$ MHz, the sampling frequency is chosen to be $4f_{IF} = 160$ MHz, as justified by the calculations below. For any other sampling frequency, a simpler solution could not have been obtained in the ideal case.

For lower frequencies, such as having the sampling frequency to be $2f_{IF} = 80$ MHz, multiplying for the Q components produces the following output:

$$\sin(\pi n) \cdot \sin(\pi n) = \frac{1}{2} [\cos(0) - \cos(2\pi n)] = \frac{1}{2} [1 - 1] = 0 \quad (14)$$

Since the Q component is completely lost, $2f_{IF}$ is not a valid sampling frequency.

For higher frequencies, such as $8f_{IF} = 320$ MHz, it can result in a better solution with less aliasing, but the FPGA cannot run at such high frequencies. Since noise levels are low, 160 MHz is employed to obtain an acceptable output, though it cannot perform as much filtering.

$$f_S = 4f_{IF} = 160 \text{ MHz} \quad (15)$$

Transitioning from continuous-time to discrete-time:

$$t = nT_S, \quad T_S = \frac{1}{f_S} = \frac{1}{4f_{IF}} \quad (16)$$

$$\cos(2\pi f_{IF} t) \Rightarrow \cos(2\pi \frac{nf_{IF}}{4f_{IF}}) = \cos(\frac{\pi}{2} n) \quad (17)$$

The I/Q mixer has a multiplier for the I component:

$$\cos(\frac{\pi}{2} n) \cdot \cos(\frac{\pi}{2} n) = \frac{1}{2} [\cos(0) + \cos(\pi n)] = \frac{1}{2} [1 + \cos(\pi n)] \quad (18)$$

From equation 18, the output of the I/Q mixer for the I component will consist of the following signal: 1, 0, 1, 0, 1, Applying the low-pass filter will result in the following signal: 1, 1, 1, 1, 1, ... , which is exactly the low frequency component of the mixed signal.

The I/Q mixer has a multiplier for the Q component:

$$\sin(\frac{\pi}{2} n) \cdot \sin(\frac{\pi}{2} n) = \frac{1}{2} [\cos(0) - \cos(\pi n)] = \frac{1}{2} [1 - \cos(\pi n)] \quad (19)$$

From equation 19, the output of the I/Q mixer for the Q component will consist of the following signal: 0, 1, 0, 1, 0, Applying the low-pass filter will result in the following signal: 0, 1, 1, 1, 1, ... , which is also exactly the low frequency component of the mixed signal after the first element.

The above calculations applies for QPSK. 16 QAM calculations are similar, except that the amplitude varies, and for the ideal case for BPSK, the Q component is equal to 0.

When there is a phase error, the output is the following for BPSK and QPSK:

$$\cos(\frac{\pi}{2} n + \theta) \cdot \cos(\frac{\pi}{2} n) = \frac{1}{2} [\cos(\theta) + \cos(\pi n + \theta)] \quad (20)$$

For 16 QAM, the equation is multiplied by the amplitude:

$$A \cos(\frac{\pi}{2} n + \theta) \cdot \cos(\frac{\pi}{2} n) = \frac{A}{2} [\cos(\theta) + \cos(\pi n + \theta)] \quad (21)$$

When there is error in the frequency, the output is the following:

$$\cos[(\frac{\pi}{2} + \rho)n] \cdot \cos(\frac{\pi}{2} n) = \frac{1}{2} [\cos(\rho n) + \cos[(\pi + \rho)n]] \quad (21)$$

Combining the two by modeling the deviation in frequency as an instantaneous phase difference as shown in Equation 22:

$$\Theta(n) = \rho n + \Theta + N(\mu, \sigma^2) \quad (22)$$

$$\cos[\frac{\pi}{2}n + \Theta(n)] \cdot \cos(\frac{\pi}{2}n) = \frac{1}{2} [\cos(\Theta(n)) + \cos[\pi n + \Theta(n)]] \quad (23)$$

where ρn is the phase error caused by the frequency difference, Θ the static phase error, and $N(\mu, \sigma^2)$ the Gaussian random variable for additive white Gaussian noise.

3.2.2 Loop Filter Design

A proportional-integral controller was chosen for the loop filter, as shown in Figure 15. The input error is the detected phase error from the phase detector. As the proportional gain (K_p) increases, the response is faster, but the system may become unstable and oscillate, while as when K_p decreases, the response is more stable, but slower. As the integral gain (K_i) increases, the system is able to track larger constant errors, but it is less responsive to new errors, while as when K_i decreases, the system is more responsive to new errors, but it can no longer track larger constant errors. Therefore, both gains offer a considerable tradeoff.

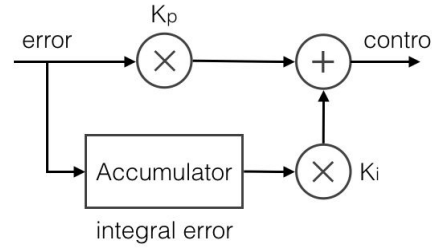


Figure 15. Block diagram showing the loop filter design

In order to take advantage of the positive aspect of both gains, the system makes use of two loop filters with different characteristics. The seeking loop filter, which is employed when no lock is detected, is chosen to have high K_p and low K_i , to track large constant errors with a fast response. On the other hand, when a lock is detected, the locked loop filter, with its low K_p and high K_i , allows a stable system that is responsive to new errors.

Figure 16 shows the constellation for the seeking loop filter (high K_p low K_i) and Figure 17 shows the constellation for the locked loop filter (low K_p high K_i). As it can be observed, the seeking loop filter rotates more. While generating these graphs, it has also been observed that the using seeking loop filter locks at the 36th cycle, while as using the locked loop filter alone locks at the 117th cycle. Note that the symbol changes every 50 cycles, which is the reason why the slower locking graphs have multiple symbols shown on the constellation.

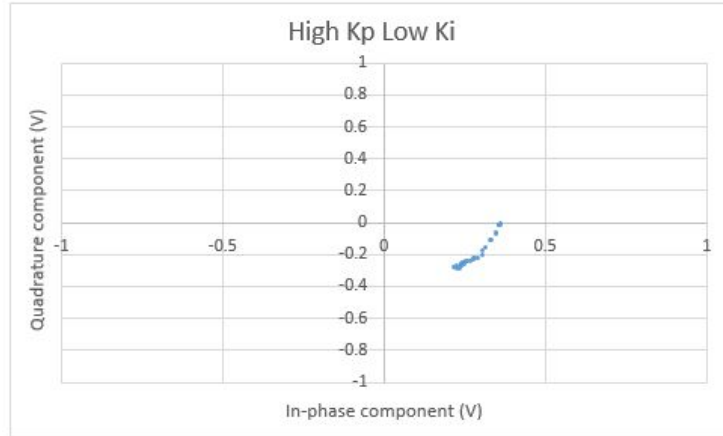


Figure 16. Constellation for the seeking loop filter (high K_p low K_i) before lock

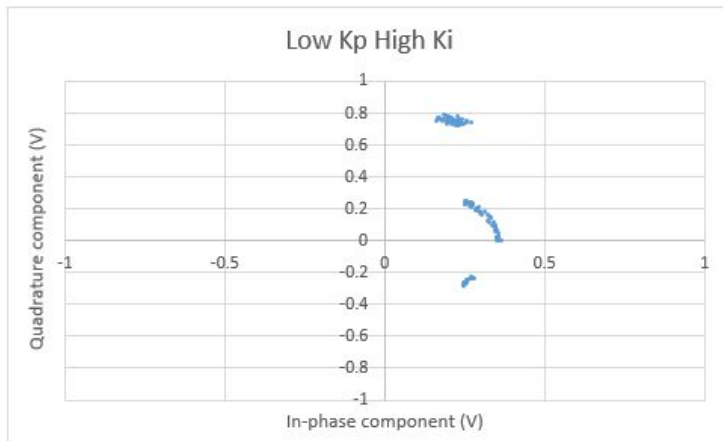


Figure 17. Constellation for the locked loop filter (low K_p high K_i) before lock

Figure 18 shows the constellation for the dual loop filter, which combines the seeking and the locked loop filters. The dual loop filter locks at the 78th cycle, which is around halfway between the lock cycles for seeking loop filter and locked loop filter.

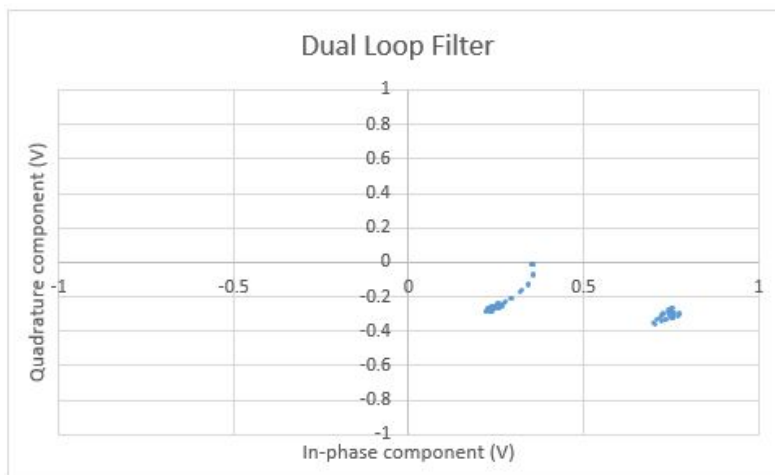


Figure 18. Constellation for the dual loop filter before lock

3.2.3 Digital Low Pass Filter Design

For a finite impulse response (FIR) filter, which is a filter that has an impulse response settling to zero in finite time as shown in Figure 19, there are many delay blocks and summers. In order to improve on the response time, as seen in Figure 20, a simple low-pass filter with one delay block is applied, thus resulting in a frequency response that involves more calculations. The ADC converter samples at 160 megasamples per second (MSPS). Since the bandwidth is at 20 MHz, according to Nyquist sampling theorem, the sampling rate is at 40 MSPS. Therefore, $160 \text{ MSPS} / 40 \text{ MSPS} = 4$ samples can be used to determine each symbol.

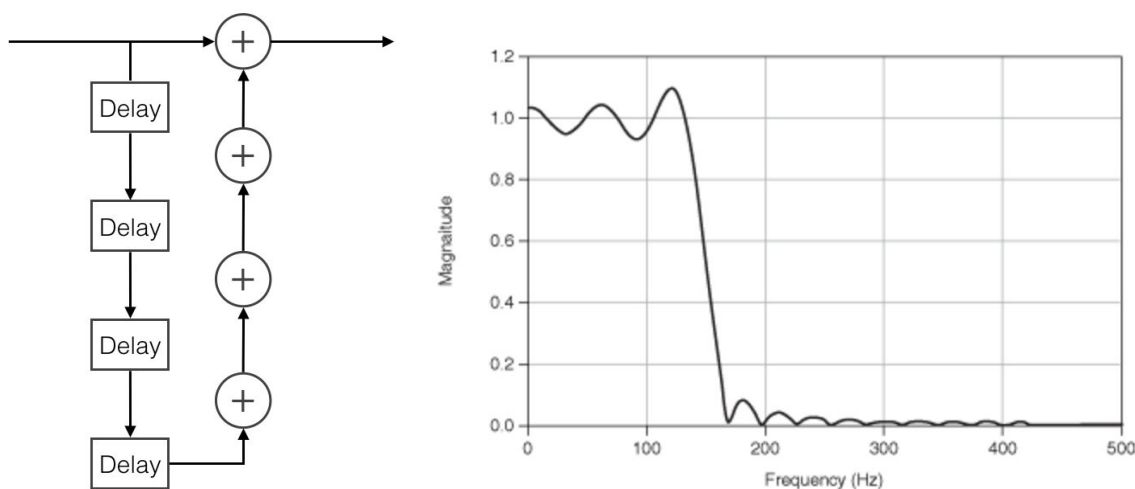


Figure 19. Block diagram and magnitude response for a simple FIR filter [14]

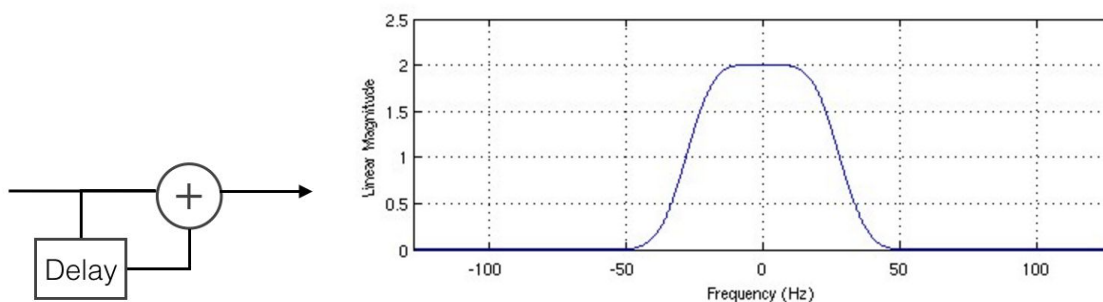


Figure 20. Block diagram and magnitude response for a simple low-pass filter [15]

3.2.4 Demodulator Performance with Noise Added

Figure 21 shows the constellation plots at various SNR ratios. At the 0dB SNR, the demodulator is incapable of locking on to the carrier frequency. At 10 dB SNR, the demodulator is able to locking on

to the carrier frequency but frequently loses sync. Acceptable performance appears when the SNR ratio is 20 dB or higher.

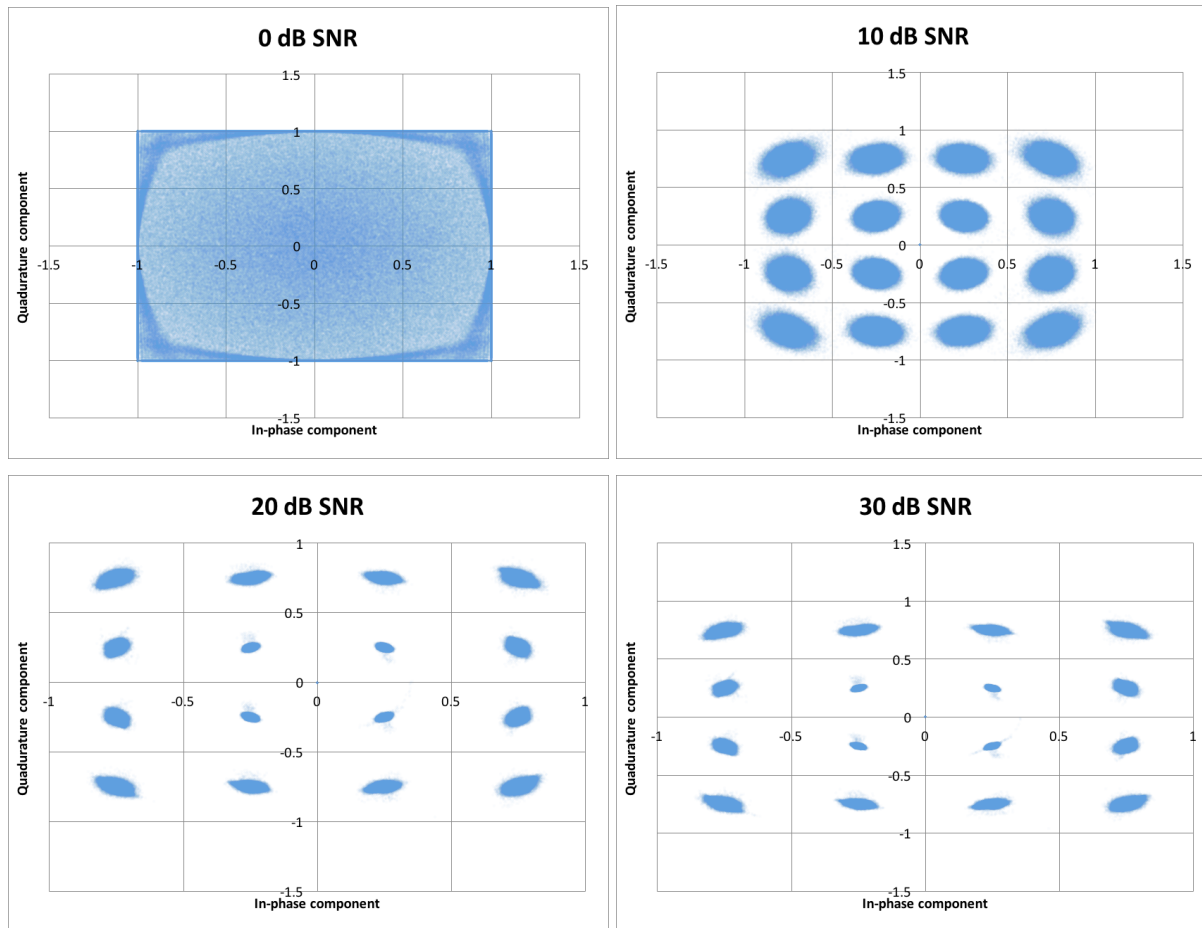


Figure 21. Constellations at 0dB, 10dB, 20dB and 30dB SNR ratios

3.2.5 Decision Device

The decision regions are regions where symbols are determined to be the ideal symbol, based on the I and Q values. The decision devices for BPSK and QPSK are trivial and will not be analyzed in detail here. The decision device for BPSK is simply detecting the sign of the I component, while the QPSK decision device detects which quadrant the symbol is in. The two options for 16 QAM decision devices are polar and rectangular, as depicted on Figures 22 and 23. Polar has been chosen over the rectangular decision device because it is able to handle rotation and amplitude changes. If the points in the rectangular decision device rotates, it is more difficult to use the symbol for phase detector as, for example, the point (0.75, 0.25) can rotate into the decision region for the point (0.75, 0.75). In the polar case, the point (0.75, 0.25) can rotate into the decision region for (0.25, 0.75) or (-0.75, 0.25) and the symbol will indeed be wrong, but with enough other symbols received, the constellation will rotate into its normal upright position.

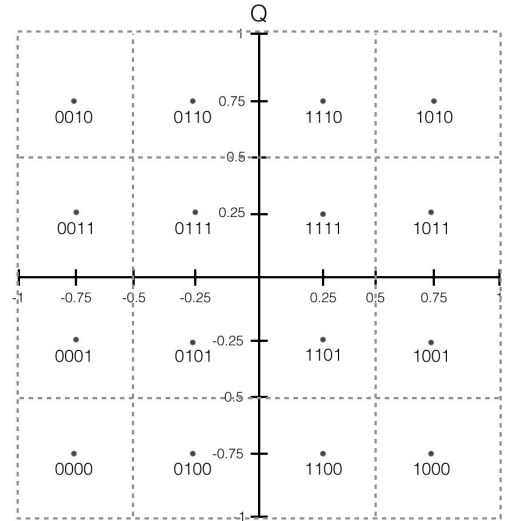


Figure 22. Rectangular decision regions with constellation

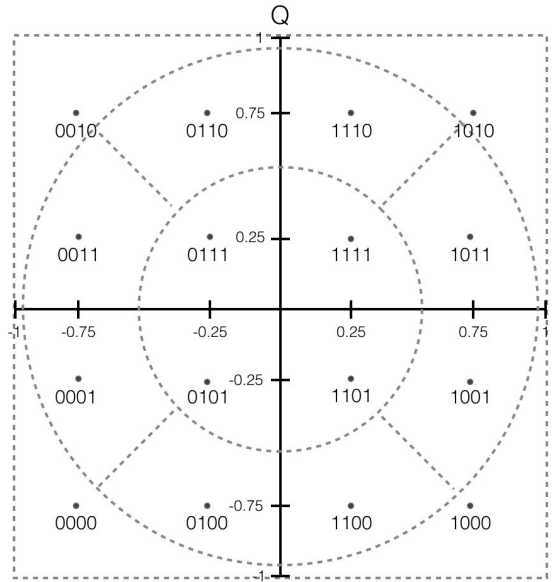


Figure 23. Polar decision regions with constellation

3.3 Subsystem 3 Design: Software Subsystem

After demodulation, the symbol is passed to the software subsystem through the communication channel. The control service is responsible for running a background service that pulls in data from the FPGA board, and pushes that information to the board database, which contains all information regarding the configurations and settings.

The user interface would allow the users to manipulate the board configuration, and to access tutorials for learning about RF communication. The web server connects to the user interface on

the client side, and communicates with the user and board database. The user database contains all user information, for both students and instructors.

3.3.1 Web Server

In terms of web server, several viable options are available, including Django, Ruby on Rails, and Node.js. As observed in Table 8, functionality is the main criterion, as the framework needs to have all the functionalities needed to build the software suite. All three options have a perfect score on this criterion because they all include the necessary tools. In terms of performance, Ruby on Rails has the worst performance, while as Django and Node.js have higher performance. Finally, the last criterion is engineering cost, which is how familiar current engineers working on this project are to the framework. Currently, Django is the fastest to implement, as members of the group have experience with this technology. Ruby on Rails gets a relatively high score as compared to Node.js because it is easier to learn. From the total score, it can be seen that Django should be selected as it has the required functionalities and good performance, and no learning time is necessary.

Table 8. Decision matrix comparing web servers

Web server	Criteria			Total
	Functionalities	Performance	Engineering cost	
	0.5	0.1	0.4	
Django	10	9	10	9.9
Ruby on Rails	10	6	8	8.8
Node.js	10	9	6	8.3

3.3.2 Database

The information that needs to be included in the database comprises of user data and board data. As previously mentioned, user data contains values the user needs to authenticate to the application, while as board data describes the various configurations for the board.

A solution has been derived where board data configurations are stored in a database table, similar to the user data, as seen on Figure 24. In user data, the 'user_type' field stores whether the user is a student or an instructor, and the 'board_data' is a string that stores the user's last board configuration. However, according to industry standards, storing configuration information field by field in a database is very inefficient. Therefore, configuration data are most often stored in JSON files, as shown in Figure 25. Since the board data is organized based on the various subsystems, JSON makes it very easy to classify the fields. The tradeoff is that the entire JSON objects needs to be rewritten after each modification.

```

CREATE TABLE user
(
    username VARCHAR(20) NOT NULL,
    password VARCHAR(20) NOT NULL,
    user_type int NULL,
    email VARCHAR(100) NOT NULL,
    board_data VARCHAR(MAX),
    PRIMARY KEY username
);

```

Figure 24. SQL code for creating the user table

```

board_data = "{
  'setup': {
    'ADC': null,
    'clock_sync': null
  },
  'RF_frontend': {
    'center_freq': null,
    'LNA': null
  },
  'IF': {
    'demodulator': null
  },
  'baseband': {
    'baseband_freq': null,
    'inv_I': null,
    'inv_Q': null,
    'swap_IQ': null,
    'demodulator': null
  },
  'other': {
    'attenuator': null,
    'noise_gen': null
  }
}"

```

Figure 25. JSON file with board data

4 DISCUSSION AND PROJECT TIMELINE

4.1 Evaluation of Final Design

The Educational RF Receiver has propensity for meeting the project design objectives and specifications. Section 3.1.3 (gain stage design) in conjunction with section 3.2.4 (demodulator performance with noise added) shows that the receiver is capable of correctly receiving and demodulating rf signals with a sensitivity of -70dBm. Section 3.2 (baseband subsystem) shows how the various demodulation schemes are implemented and how the 20MHz bandwidth signal can be demodulated. Section 3.1 (RF/IF subsystem) discusses how signals between 0.8GHz and 1.9GHz can be mixed down to 40MHz to be demodulated. Components will be selected that can attenuate between 30dB and 70dB as well as a directional coupler which directs -10dB of signal for measurement. There will also be a design to allow the connection of an additive noise source and power protections. The webserver design as in section 3.3 (software subsystem) will all the simultaneous connection of 30

users who can modify settings with a latency below that of 3s and support web browsers Safari, Google Chrome and Firefox. The software UI design is being planned carefully with usability and low learning curve in mind so that all features are active in a intuitive package that can be accessible within 4 clicks. Not every circuit component has been finalized yet but once the PCB is designed, every effort will be made to cut costs and reduce the RF front-end component cost to below \$500. Due to RF front end components being relatively low power circuit elements, there should not be any problems reducing the power consumption of the entire system below 10W. After the PCB design is completed, insulated case will be designed that will provide adequate cooling. Overall, the detailed design shows that the product is not only feasible but should be able to meet all defined specifications.

4.2 Use of Advanced Knowledge

The design of the RF Educational Receiver uses knowledge from a variety of upper year courses. Each of the subsystems used knowledge from at least 2 upper year courses.

The RF and IF subsystems uses upper year knowledge from ECE473/ECE671 in order to design and analyse microwave circuits as well as custom image filters. In order to understand different radio architecture and complete radio link calculations, knowledge from ECE432 was necessary. In order study the gain stage design and perform noise figure calculations, knowledge from ECE 474 was invaluable. All of these 4th courses stem from 3rd year courses ECE318 and ECE375.

The baseband subsystem relies heavily on FPGA design. ECE327 and ECE429 labs/assignments have a very heavy FPGA design component that is necessary to design the FPGA logic for the demodulator and decoder. Sampling and digital filter design is taught in ECE413 for digital signal processing. ECE380 teaches controller design which was important in designing both the seeking and locked loop filters. Again, ECE318 was useful here in order to design the different demodulation schemes: BPSK, QPSK and 16 QAM. On the software side, ECE358 is important to design the packet format to communicate over the network and ECE356 knowledge is necessary to design the database housing user information.

4.3 Creativity, Novelty, Elegance

The most creative parts of the receiver design is linked to its educational value. One of the features that was implemented early on in the design process is the capability to attach an attenuator + noise generator at the input of the receiver. These are not components present on a normal receiver but the reason they were added was so that students can simulate noise and attenuation as one would experience when accepting a signal from an antenna in real life applications. Another feature that would be very valuable to students is our decision to include a bypass option where an image filter is not connected in the design. This helps show the value of the image filter and how the signal is distorted without it.

Another creative decision was to implement a digital IF domain. There are some inherent advantages to working in digital as opposed to analog. Digital is more flexible than analog and changes have a much shorter lead time, especially minor changes to quickly do some more tuning. Furthermore, while perfect analog components do not exist, ideal components can be created in digital without any noise or imperfections. In the digital design, 2 control loops were created for the loop filter. As a result, the benefits of both can be applied to the product.

4.4 Student Hours

The total number of hours logged by each student as of June 30, 2016 follows Table 9 below.

Table 9. Hours Logged by Students as of June 30, 2016

Student	Total Hours Logged
[REDACTED]	97
[REDACTED]	87
[REDACTED]	91
[REDACTED]	133
[REDACTED]	166.5

4.5 Potential Safety Hazards

The receiver will have a enclosed box designed for the final demonstration as safety protocol of all electrical circuit design. During designing stage, safety hazards could potentially occur similar to other electrical projects.

The receiver does not require high voltage therefore there is less risk for electrical shock. With any electrical circuits, it could run the risk of shorted connections which can heat up drastically and may burn. To mitigate the issue a fuse will be designed onto the circuit as well as ensure precautions are taken before working on the circuit board. The person who's working on the circuit board will not be wearing metal rings and tie long hair back.

Will be working on soldering at designated labs to use the correct tools and work under safe environment. Ensure first aid supply (such as band aid) is available in case of accidents. Soldering could result in solder splatter and burn from hot soldering irons. Proper equipment such as fume extractor will be used and protective gear and goggles will be worn while working with hazardous equipment.

4.6 Project Timeline

The project timeline for both ECE498A and ECE498B loads is shown in Figure 26.

Educational RF Receiver

Jun 29, 2016

Gantt Chart

3

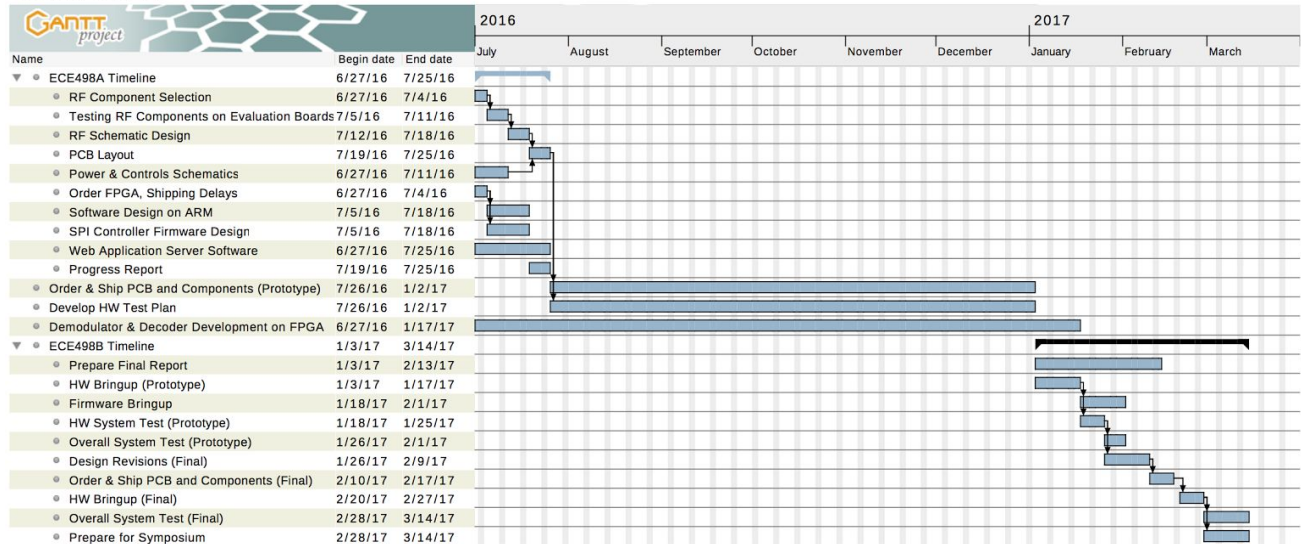


Figure 26. ECE498A and ECE498B Project Timeline

5 RISK ASSESSMENT

The major risks identified are detailed in Table 10 and include hardware acquisition, technical risks, and group member availability and competency.

Table 10. Assessment of Potential Risks Barring Project Completion

Risk	Impact	Likelihood	Mitigation action
Development time for FPGA can be very long in the case that the scope is set too large	High	High	All algorithms will be implemented in sequential code on a CPU first which, albeit having much slower performance, will have a much reduced development time compared to that of FPGA. This way, at least a functional prototype can be obtained at the end.
Not enough IO pins, may require baseband design change last minute, cannot lock into a particular development board	Medium	Low	A development board with a high-speed I/O connector as well as at least 32 low speed I/O pins will be chosen. Configurable components will be selected to adhere to a single bus protocol to minimize the number of pins required.

			Additionally, it is possible to add microcontrollers if there really is not enough I/O pins after taking the aforementioned steps.
Insufficient number of hardware revisions possible: there is only time to compete board 1 revision before delivering working prototype.	High	Medium	In order to learn as much from the first revision as possible and require as little changes as possible for the final revisions, RF simulations will be run using industry software to perfect board design as much as possible.
There might be issues getting hardware supplies quickly. Components can also become obsolete without notice.	Medium	Low	The project will try to order from mostly local suppliers. Any components bought to build the first revision will be bought in enough quantities to also support the second revision.
Signal leakage/jamming	High	Medium	There is a high risk of digital/RF/power circuitry leaking unto RF traces. The project will explore using stripline traces as well as construct a comprehensive test plan to account for all possible use cases.
Team Dynamics	Medium	Medium	There is always a risk of conflict between group members in high stress situations. The project had a team building event at the beginning of the 4A term and will have additional events throughout the lifetime of the project
High RF/baseband decode learning curve	Medium	High	The project team was aware that the project is rather technically challenging and had already started to work on research for all components related to the receiver design during the work term before the 4A term.

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